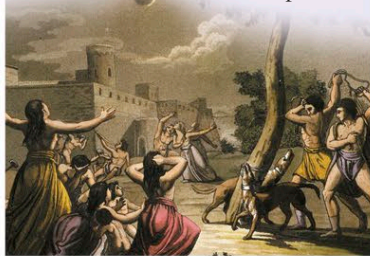


MYTHS AND STORIES

EVIL PORTENTS

Astronomers have predicted eclipses reliably since about 700 BCE, but that has not stopped doomsayers and astrologers from reading evil omens into these routine celestial events. They have often prophesied disasters associated with eclipses, and although they meet with no more than occasional success, some people listen. The Incas below, for instance, are pictured as awestruck by an eclipse, in a European atlas of 1827. Eclipses may not be useful for predicting the future, but accounts of past eclipses are of great value to today's historians, who can calculate the dates of events with great precision if the historical accounts include records of eclipses.



LUNAR ECLIPSES

As the Moon orbits the Earth, it occasionally moves into Earth's shadow—an occurrence called a lunar eclipse—or blocks sunlight from reaching a part of Earth's surface—a solar eclipse. Eclipses do not happen every month, because the plane of the Moon's orbit around Earth does not coincide with the plane of Earth's orbit around the Sun. Nevertheless, an eclipse of some kind occurs several times each year. Lunar eclipses are common, occurring two or three times a year, always during a full moon. Astronomers classify lunar eclipses into three different types. In a penumbral eclipse, the Moon passes through Earth's penumbra (part-shadow), leading to only a slight dimming of the Moon. In a partial eclipse, a portion of the Moon passes through Earth's umbra (full shadow), while in a total eclipse the whole Moon passes through the umbra.

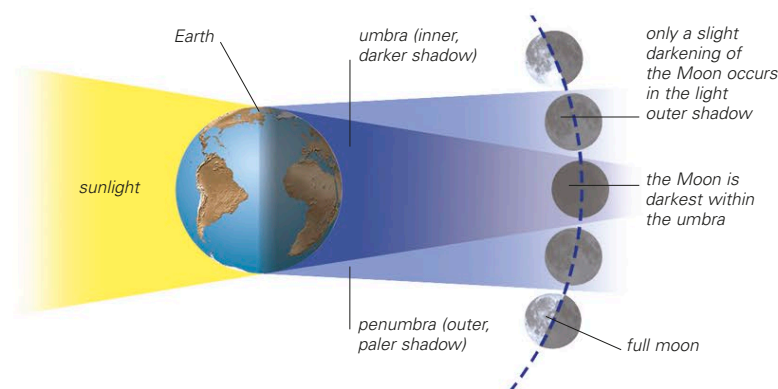
MECHANICS OF A LUNAR ECLIPSE

Earth's shadow consists of the penumbra, within which some sunlight is blocked out, and the umbra, or full shadow. In a total eclipse, the Moon passes through the penumbra, the umbra, and then the penumbra again.



TOTAL LUNAR ECLIPSE

This composite photograph shows stages of a total lunar eclipse. The moon appears red at the eclipse's peak (bottom left), because a little red light is bent toward it by refraction in Earth's atmosphere.



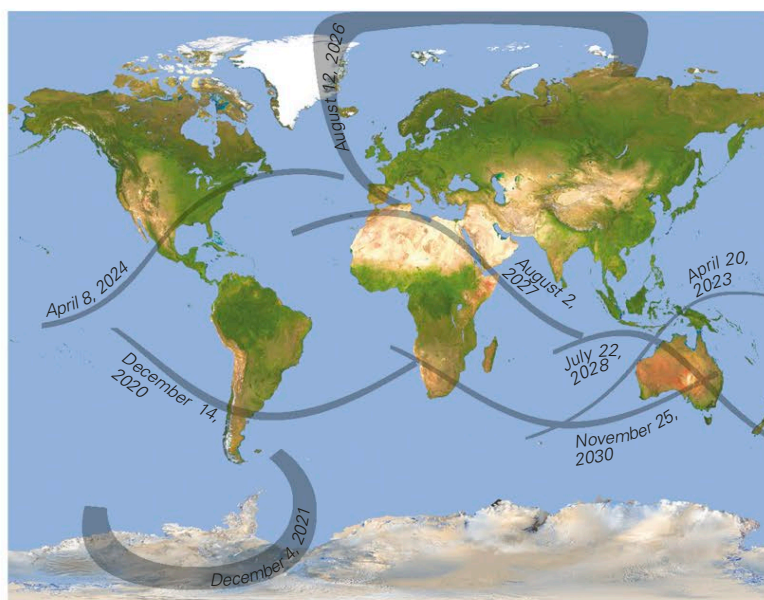
SOLAR ECLIPSES

An eclipse of the Sun occurs when the Moon blocks sunlight from reaching part of the Earth. During a total eclipse, viewers within a strip of Earth's surface, called the path of totality, witness the Sun totally obscured for a few moments by the Moon. Outside this area is a larger region where viewers see the Sun only partly obscured. More common are partial eclipses, which cause no path of totality. A third type of solar

TOTALITY PATHS

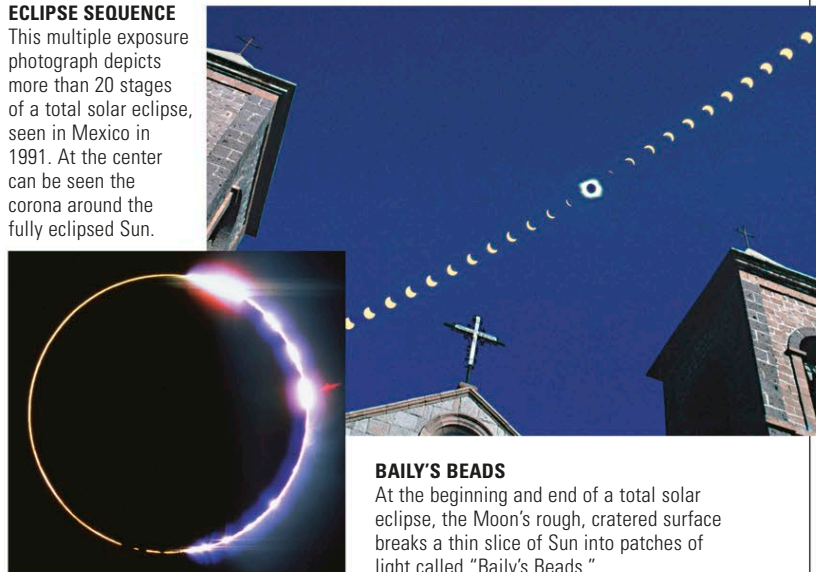
The part of Earth's surface over which the Moon's full shadow will sweep during a total solar eclipse, called the path of totality, can be predicted precisely. Below are the paths for eclipses up to 2030.

eclipse is the annular eclipse, occurring when the Moon is farther from Earth than average, so that its disk is too small to cover the Sun's disk totally. At the peak of an annular eclipse, the Moon looks like a dark disk inside a narrow ring of sunlight. Solar eclipses happen two or three times a year, but total eclipses occur only about once every 18 months. During the period of totality, the Sun's corona (its hot outer atmosphere) becomes visible.



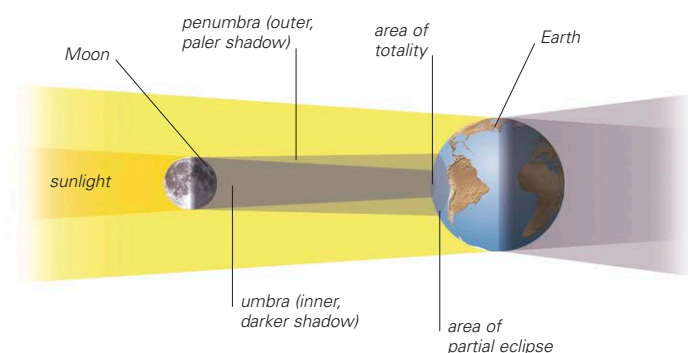
ECLIPSE SEQUENCE

This multiple exposure photograph depicts more than 20 stages of a total solar eclipse, seen in Mexico in 1991. At the center can be seen the corona around the fully eclipsed Sun.



BAILY'S BEADS

At the beginning and end of a total solar eclipse, the Moon's rough, cratered surface breaks a thin slice of Sun into patches of light called "Baily's Beads."



MOON SHADOW

The shadow cast by the Moon during a total solar eclipse consists of the central umbra (associated with the area of totality) and the penumbra (area of partial eclipse).